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Regional-scale digital soil mapping of key GlobalSoilMap properties using machine learning and MODIS data in South Africa

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ABSTRACT

Digital soil nutrients (DSN) information plays a critical role in enhancing management of different ecological and agricultural systems. Mapping soil nutrient parameters at regional scale countries like South Africa that rely hugely on agricultural production but lack DSN datasets products that can enhance usage and accessibility of such information products to a wider range of communities. This study aims to contribute towards the GlobalSoilMap initiative by utilizing the South African Soil Database and MODIS datasets, to analyze and map selected soil nutrients using the Quantile Regression Forest model across South Africa. The soil organic carbon, pH-soil, cation exchange capacity (CEC), clay, sand, and silt parameters are mapped at 0–30 cm depth, including their uncertainties. The local DSM were comparable to the iSDA products with correlations above 80 % for selected soil parameters which proves that the country level DSM product can provide locally relevant information than global products. Variable importance showed that vegetation indices, land surface temperature and precipitation have a strong relationship with soil properties. Our results will empower policymakers to incorporate soil nutrient distribution information in their decision-making processes and farmers can use the same for improved soil management practices.

1. Introduction

Soil health is critical for sustainable agriculture, biodiversity and ecosystem services maintenance, and generally involves a significant storage for carbon. It is, inarguably, the most vital natural resource, with the potential to negatively affect many socio-economic and environmental aspects worldwide if damaged. In the age of global change, preserving the soil and preventing its loss through erosion and degradation by unsustainable agricultural management practices is more crucial than ever. Essentially, the production of sufficient food and achievement of global food security imperatives such as the United Nations Sustainable Development Goals (SDGs) and Land Degradation Neutrality (LDN) hinges on sustaining the quality and productivity of the soil. For example, achieving SDG goal 2, 6, 13, and 15 which aims to reduce the impact of weather conditions, restore degraded lands, achieve equitable, affordable, and sustainable food production can only be realised with sustainable soil management practices.

Levels of soil chemical properties such as organic carbon (OC), potassium (K), magnesium (Mg), sodium (Na), pH, cation exchange capacity (CEC), electrical conductivity (EC), calcium (Ca), phosphorus (P), zinc (Zn), and nitrogen (N), are some of the significant 42 indicators of soil quality (Séré et al., 2024). For example, characterizing soil chemical properties is critical for understanding available soil nutrient levels at any given time during crop development, allowing farmers and land managers to make informed decisions about fertilizer application, and crop selection, as well as informing overall land management (Marshall and Thenkabail, 2015; Nduku et al., 2024). Moreover, characterizing soil physical properties such as texture, e.g., sand (particle size: 0.05–2 mm), silt (particle size: 0.002–0.05 mm), and clay (particle size: 0.002 mm), is equally critical since they affect water retention, drainage, and nutrient availability (Liu et al., 2015; Castaldi et al., 2016). Soil texture influences the selection of suitable crops and irrigation methods, the structure and diversity of the bacterial community, and helps in soil management and conservation practices (Duan et al., 2022; Siqueira

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et al., 2023).

Regional scale soil characterization is still a challenging task as field and lab-based methods are only limited to certain spatial and temporal scales and are both expensive and labour intensive (Rossel et al., 2006; Soriano-Disla et al., 2014). Moreover, technologies such as MIR Spectroscopy and Remotely Piloted Aerial Systems (RPAS, commonly UAVs), albeit being accurate, are costly, hence in accessible and are mostly site-specific (Alomar et al., 2022; Kganyago et al., 2024). Therefore, satellite earth observation technology is a viable alternative to accurately characterize the spatial variations of soil quality at larger scales, to support policy and decision-making. For this, a myriad of satellites exists and provide time-rich and spatially explicit data, providing prospects for improving existing knowledge and information extraction relevant to soil quality preservation.

Indeed, the accuracy of the digital soil maps relies on the analytical techniques employed with the use of machine learning algorithms shown to yield good results (Abd El-Hamid et al., 2023; Kebonye et al., 2023). Some of the dominantly applied algorithms documented in literature are Partial Least Squares Regression (PLSR), Support Vector Regression (SVR), Random Forest Regressing (RFR), Adaptive Boosting (AdaBoost), Gradient Boosting Machine (GBM), Neural Networks (NN) and Extreme Gradient Boosting (XGB) (Datta et al., 2023). Furthermore, Landsat data is mostly used due to its long-term archives of medium spatial resolution data (Vågen et al., 2013; da Silva Chagas et al., 2016; Zeng et al., 2020; Novais et al., 2021; Abd El-Hamid et al., 2023; Shirazi et al., 2023; Kebonye et al., 2023). However, for certain applications, such as national, sub-national, and regional scales, soil mapping for applications that require less precision, e.g., policymaking and implementation and land resource planning and management, the burden of DSM using big data can be avoided by adopting relatively lower resolution data from Moderate-resolution Imaging Spectroradiometer (MODIS). Vågen et al. (2016) show that 96 coarser scale DSM from 250 m to 1000 m MODIS can serve as prior estimates for mapping at local scales.

Well-established datasets from the MODIS are critical to achieving the DSM, particularly owing to their long-term series and relatively high spatial resolution, i.e., up to 250 m, compared to Advanced Very High-Resolution Radiometer (AVHRR). Previous research efforts such as Zhang et al. (2022) and Vågen et al. (2016) demonstrate the importance of MODIS-derived products and have culminated in soil data products such as the GlobalSoilMap (GSM, [a global soil properties map at 100-m resolution]) (Hempel et al., 2014), which is dubbed as one of the pioneering projects for DSM, is part of the Global Soil Partnership (GSP). A recent product, SoilGrids 2.0 (Poggio et al., 2021), has global coverage and a spatial resolution of 250 m and was generated with Quantile Random Forest Algorithm (QRF). This demonstrates its robustness for operational digital soil mapping to support land use planning, agricultural management, and soil conservation. Although these initiatives exist, there is a need for improved DSM to support sustainable land use planning, continuous monitoring of soil health, helping to identify areas at risk of degradation early on and empowering local farmers and communities with the knowledge they need to improve their agricultural practices. Moreover, it is critical to continue developing and testing the performance of QRF under regional contexts. The QFR model quantifies uncertainty by generating prediction intervals from conditional quantiles, which is particularly advantageous in mapping soil property distribution along with their associated uncertainties. Other models such as Random Forest and Partial Least squares regression are limited in this respect. Random forest only provides point predictions without uncertainty measures and partial least squares is inefficient in modelling the nonlinear relationships present in complex soil data (Coulston et al., 2016; da Silva Chagas et al., 2016; Takoutsing and Heuvelink, 2022).

Numerous global DSM products are available, however, their main disadvantage is that they lack the resolution needed for national and regional applications. Traditional methods focusing on soil surveys and lab analyses are costly and limited in scale. Machine learning models such as QFR remain underutilised in local contexts. Therefore, there is a need for a local DSM which supports sustainable land management and supports policy. The specific objectives of the study were: 1) to identify influential soil parameter-specific environmental parameters critical for DSM. 2) To spatially characterize various soil parameters at national scale using a robust QRF model and 3) to compare the locally generated DSM with existing soil databases.

2. Materials and methods

2.1. South African soil legacy database

The Soil African national soil database was developed and managed by the Agricultural Research Council (ARC). The database contains layer-wise soil property data for over 10,000 georeferenced (latitude and longitude) soil profiles distributed across the country (Fig. 1). The soil data were collected as part of the national soil survey project and the database continues to be updated. Primarily, the soil samples were collected from different land use types (e.g., farming, grazing land). The soil profile samples within the database range from 0 to 120 cm and it has about 24 variables (macro, micro and texture information). Table 1 below provides the description of soil datasets selected in this study using domain knowledge on Digital soil mapping methodological framework. The number of points for soil variables varied from 8307 to 10,113. The selected soil properties in Table 1 were harmonized to produce a synthetic horizon layer of 0–30 cm tillage zone which is important for assessing chemical and physical properties in relation to agriculture that relies on top soil (Olson and Al-Kaisi., 2015). This was done by using a mass preserving spline function fitted to each individual soil profile (Bishop et al., 1999).

2.2. Environmental covariates

Different data sources representing major soil-farming factors were considered as environmental covariates (Appendix A: Table A1). The datasets had varying spatial resolutions ranging from 250 m to 1000 m. A bilinear spatial resampling method was used in Google Earth Engine to interpolate the data to 250 m spatial resolution (Gorelick et al., 2017). Different statistical variables were derived from the dataset such as the minimum, maximum, mean and standard deviation of e.g., Normalized Difference Vegetation Index (NDVI), precipitation, Potential Evapotranspiration (PET), and temperature. A total of 72 covariates were generated as input into the Quantile Regression Forest (QRF) model.

2.3. Quantile random forest (QRF) model

Quantile Regression Forest model is a generalized non-parametric, tree-based ensemble method for estimating conditional quantiles (Meinshausen and Ridgeway, 2006). It was derived from the well-known Random Forest Model, originally proposed by Breiman (2001). The QRF employs the weighted empirical distribution of training labels to obtain the predictive distribution, while the Random Forest model outputs the weighted mean value of training labels (Johnson, 2024). This unique feature allows QRF to also outputs probabilistic predictions for regression problems, which are necessary in constructing uncertainty estimates (Petropoulos et al., 2022).

Formulation of the QRF as given in (Meinshausen and Ridgeway, 2006) is as follow: let X be a predictor variable and Y be a response variable. The goal is to estimate the conditional mean given by the conditional quantile function Q_τ.

$$Q_\tau(y|X = x) \text{for} \tau \in (0, 1). \quad (1)$$

The weighting function $w_i(x)$in QRF is defined as:

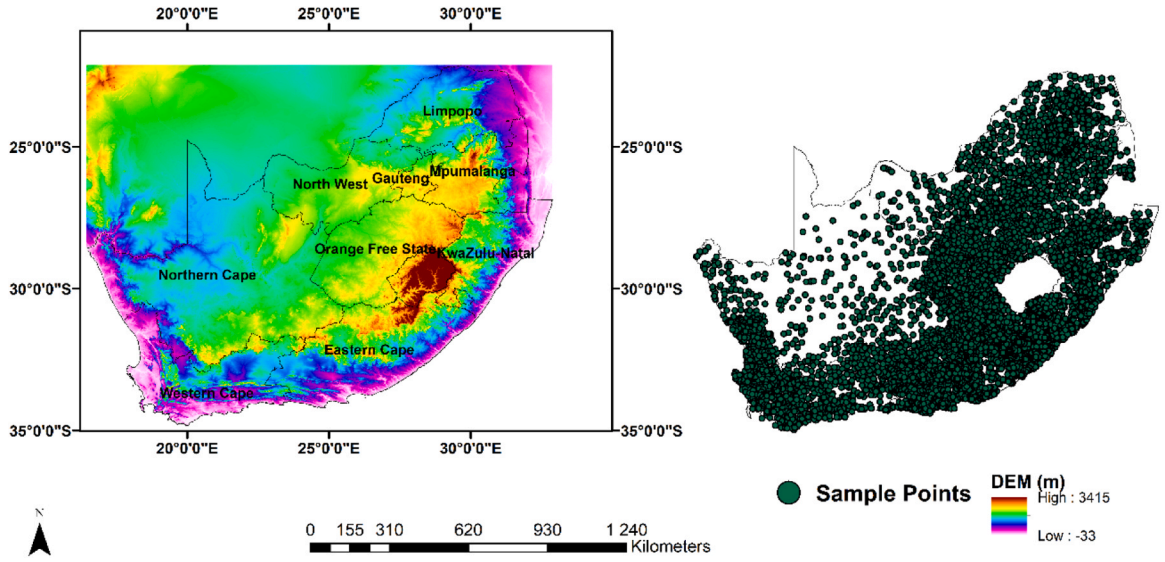


Fig. 1. Study area map illustrating (left) Digital Elevation Model (DEM) and (right) sample points.

Table 1
List of soil properties and their description used in the study. The profiles were harmonized to 0–30 cm layers.

Soil properties	Variables	Number of points	Unit	Methods	References
Carbon	SOC	8307	%	Walkley Black	Nelson and Summers (1982)
Cation Exchange Capacity	CEC	9508	cmol (+)/kg	Ammonium Acetate	Rhoades, (1982)
pH-soil	pH	8010	None	Soil to water ratios(1:2.5)	Peech, (1965); Sonmez et al. (2008)
Clay	Clay	8768	%	Bouyoucos/ hydrometer method	Gee and Bauder, (1986); Non-Affiliated Soil Analysis Work Committee (1990)
Sand	Sand	10,113	%	Bouyoucos/ hydrometer method	Gee and Bauder, (1986); Non-Affiliated Soil Analysis Work Committee (1990)
Silt	Silt	10,023	%	Bouyoucos/ hydrometer method	Gee and Bauder, (1986); Non-Affiliated Soil Analysis Work Committee (1990)

$$w_i(x) = \frac{1}{T}\sum_{t=1}^{T}\frac{I\{i \in I_t(x)\}}{|I_t(x)|}, \tag{2}$$

Where$I\{\cdot\}$is an indicator function and $|I_t(x)|$is number of training points in the leaf and these weights sum to 1. The weighted training responses are used to build the empirical conditional distribution function and the conditional distribution of Y given$X = x$ is approximated by:

$$\widehat{F}(y|x) = \sum_{i=1}^{n} w_i(x)\cdot 1\{y_i \le y\}. \tag{3}$$

The conditional quantile estimation is estimated by:

$$\widehat{Q}_\tau(x) = \inf\{y : \widehat{F}(y|x) \ge \tau\}. \tag{4}$$

The model was evaluated using a 10-fold cross validation approach repeated 10 times. From these repetitions, the best model is selected to make predictions (mean and standard deviation) of the target soil properties in Table 1. This was implemented in R-software using the Caret package (Kuhn, 2024).

2.4. Evaluation

The uncertainty assessment of the products generated in this report provides ways in which data can be interpreted. Each product is accompanied by the standard deviation map, which indicates spatial distribution of the errors. This is particularly important for understanding areas that have high and low errors. Each map also has an overall accuracy measure, which assists the user in the interpretation of the map products. The average errors are calculated between the prediction $(\widehat{z})$ and observed(z) values of soil properties S at the location i. The error ϵ is expressed using the following Eq. (5).

$$\varepsilon(S_i) = z(S_t) - \widehat{z}(S_t). \tag{5}$$

The prediction bias of the model is estimated by mean error (ME)Eq. (6) below. The negative values indicate that the predicted values are biased towards lower predictions and positive ME indicates a bias of the model towards higher predictions.

$$ME = \frac{1}{N}\sum_{i=1}^{N}\varepsilon(S_i). \tag{6}$$

The mean absolute error (MAE) Eq. (7) also measures the overall magnitude of the prediction error. The MAE values closer to 0 indicate higher accuracy of the prediction model.

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|\varepsilon(S_i)|. \tag{7}$$

The root mean square error ($RMSE$) Eq. (8) is the square root of the mean of the square of all the errors. It measures the prediction errors, and it is a good comparison parameter of different model performance since it is scale dependent.

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\varepsilon(S_i)^2}. \tag{8}$$

The modelling efficiency coefficient (MEC) Eq. (9) accounts for the

proportion of variance that is explained by a model (Janssen and Heuberger, 1995). MEC value of 1 indicates perfect scenario and 0 value means that the model does not predict the values better than the mean of the observed values would. Negative *MECs* indicate that the model predicts the values worse than the mean of the observed values.

$$MEC = 1 - \frac{\sum_{i=1}^{N} (z(S_i) - \widehat{z}(S_i))^2}{\sum_{i=1}^{N} (z(S_i) - \overline{z}(S_i))^2}. \quad (9)$$

The coefficient of determination (R^2) Eq. (10) also measures the overall accuracy of the model. The value of 1 indicates total correlation between predicted and observed values, whereas 0 indicates no correlation.

$$r^2 = \frac{\sum_{i=1}^{N} (z(S_i) - \overline{z})(\widehat{z}(S_i) - \overline{z})}{\sqrt{\sum_{i=1}^{N} (z(S_i) - \overline{z})^2} \sqrt{(\widehat{z}(S_i) - \widehat{z})^2}}. \quad (10)$$

3. Results

3.1. Distribution characteristics of SOC, pH-soil and soil mechanical composition

Univariate statistics of the soil nutrient parameters are listed in Table 2. The SOC has a mean of 1.21 % with a maximum of 5.84 %. The clay, sand and silt have the means 21.84 %, 50.04 %, and 16.57 % respectively, while the pH-soil and CEC have the mean values of 6.67 cmol(+)/kg and 10.62 % respectively. Sand exhibited highest variability of 27.84 % compared to silt at 12.2 % and clay at 14.23 %. Fig. 2 depicts the density distribution of the soil nutrient parameters. The SOC, CEC, silt and clay have similar distribution, all skewed towards the right. The pH-soil shows bimodal distribution, with two peaks occurring at about pH values of 6 and 8.5. Sand parameter shows left skewed distribution.

3.2. Uncertainty of model performance and environment factors affecting soil properties

The performance summary of the QRF model is given in Table 3 and scatter plots in Fig. 3. The model information criteria are described by ME, MAE, RMSE, R^2 and MEC. The QRF model attained varied performances across all soil variables evaluated in the study. The SOC had −0.02 ME, 0.46 MAE, 0.66 RMSE, R^2 of 0.55 and MEC value of 0.55. Similar model performance was obtained for the pH-soil with −0.01 ME, 0.54 MAE, 0.69 RMSE, R^2 of 0.61 and MEC value of 0.61. CEC showed lower performance with an R^2 value of 0.36 and RMSE of 5.63. The CEC lower performance cannot be generalized. However, it can be attributed to soil heterogeneity across the entire study area and complexities associated with RF model to predict CEC accurately using MODIS datasets. This study concluded that the best prediction accuracy explained 55 % and 61 % of the soil variability in SOC and pH-soil content, respectively. Zeraatpisheh et al. (2019) achieved similar results of 55 % soil variability for SOC. According to lower RMSE results, it could be concluded that the best performing model does not necessarily produce utmost accurate predictions. The model R^2 results attained in this study could be used as the baseline information to intensify more denser sampling in the Karoo areas. More soil sampling to update legacy datasets probably will increase model accuracy performances which is key for policymaking in regional-scale digital soil mapping.

Figs. 4 and 5 illustrate the ranked importance of each variable that was used by the QRF model to model selected soil parameters. The mean NDVI was the highest ranked variable for predicting SOC, CEC and clay parameters. The land surface temperature (LST) mean data and mean precipitation data (bio12, bio13, and bio16) were among variables that were very important in predicting pH-soil parameter. For the sand parameter, the derived evapotranspiration (maximum) and precipitation (mean of the driest quarter) parameters were important, while, for silt, mean precipitation of the driest quarter (bio17) and mean precipitation of the driest months (bio14) were the top two important variables.

Table 2
Summarizes the data products, and univariate statistics for the input data used for generating a national GSNmap product.

Variables	mean	sd*	median	min	max	range
SOC (%)	1.21	0.98	0.9	0.1	5.84	5.74
Clay (%)	21.84	14.23	18.44	1.37	64.71	63.33
Sand (%)	50.04	27.84	54.86	0	100.2	104.7
Silt (%)	16.57	12.2	15.26	0	51.18	51.18
pH-soil	6.67	1.11	6.5	4.53	9.2	4.67
CEC (cmol(+)/kg)	10.62	7	8.99	1.33	40.96	39.63

* sd: standard deviation

3.3. Spatial distribution of key GlobalSoilMap properties and innovative solutions for decision agriculture validation products

The spatial predictions of various variables of interest are depicted in Figs. 6–11, with their associated standard deviations as predicted by the QRF model in Table 3, respectively. The arid zones in South Africa i.e., Kalahari, Succulent Karoo, and Nama Karoo in the north-western part of the country, which are characterized by low rainfall, show distinct differences in chemical distribution compared to the areas that have higher rain-fall and dense vegetation cover in the north-eastern parts of the country and towards the south. Higher values of SOC are concentrated in areas that have higher rainfall and vegetation. Low SOC values are observed in the semi-arid areas. Similar and consistent observations are made for clay content in Fig. 9. Higher pH values (6−9) describing neutral soils in semi-arid areas are consistent with our expectations as these areas are characterized by high sand distribution (Fig. 10). The ability of the soils to supply soil nutrients is measured by the CEC in Fig. 7, where CEC distribution is consistent with expected fertile soils distributions as illustrated by the SOC, pH, and clay variables in Figs. 6, 8 and 9, respectively.

Our results are compared with the results from the Innovative Solutions for Decision Agriculture Ltd (iSDA) (Hengl et al., 2021). We generated 2000 random points based on the South African boundary shapefile to ensure that most areas were sampled in QGIS 3.14. The points were used to extract raster values of both ARC and iSDA raster layers. The results are depicted in Fig. 12 for sand, clay, silt and pH parameters. They indicate acceptable similarities for both low and high concentration levels of the selected parameters. The correlation between the two products was observed to be above 80 % for all the parameters compared. Lemercier et al. (2022) observed differences between the GlobalSoilMap products for France and the SoilGrids 2.0 products. Venter et al. (2021) also noticed differences in SOC between the AfSoilGrids 250 m and Harmonized World Soil Database (HWSD) over South African region. Discrepancies exist within different products due to different covariates that are used within the model, the choice of regression model used e.g., kriging or machine learning, and the scale at which the products were generated, among other factors.

4. Discussion

In this study, topsoil (0–30 cm) and key *GlobalSoilMap* properties (SOC, pH-soil, CEC, sand, clay and silt) were mapped using the South African national soil database that is managed by the Agricultural Research Council (ARC) to support the *GlobalSoilMap* project (https://www.fao.org/global-soil-partnership/en/). There is a need to evaluate local DSM products on a local scale. Such initiatives enhance the concept of soil security (Koch et al., 2013; McBratney et al., 2014;

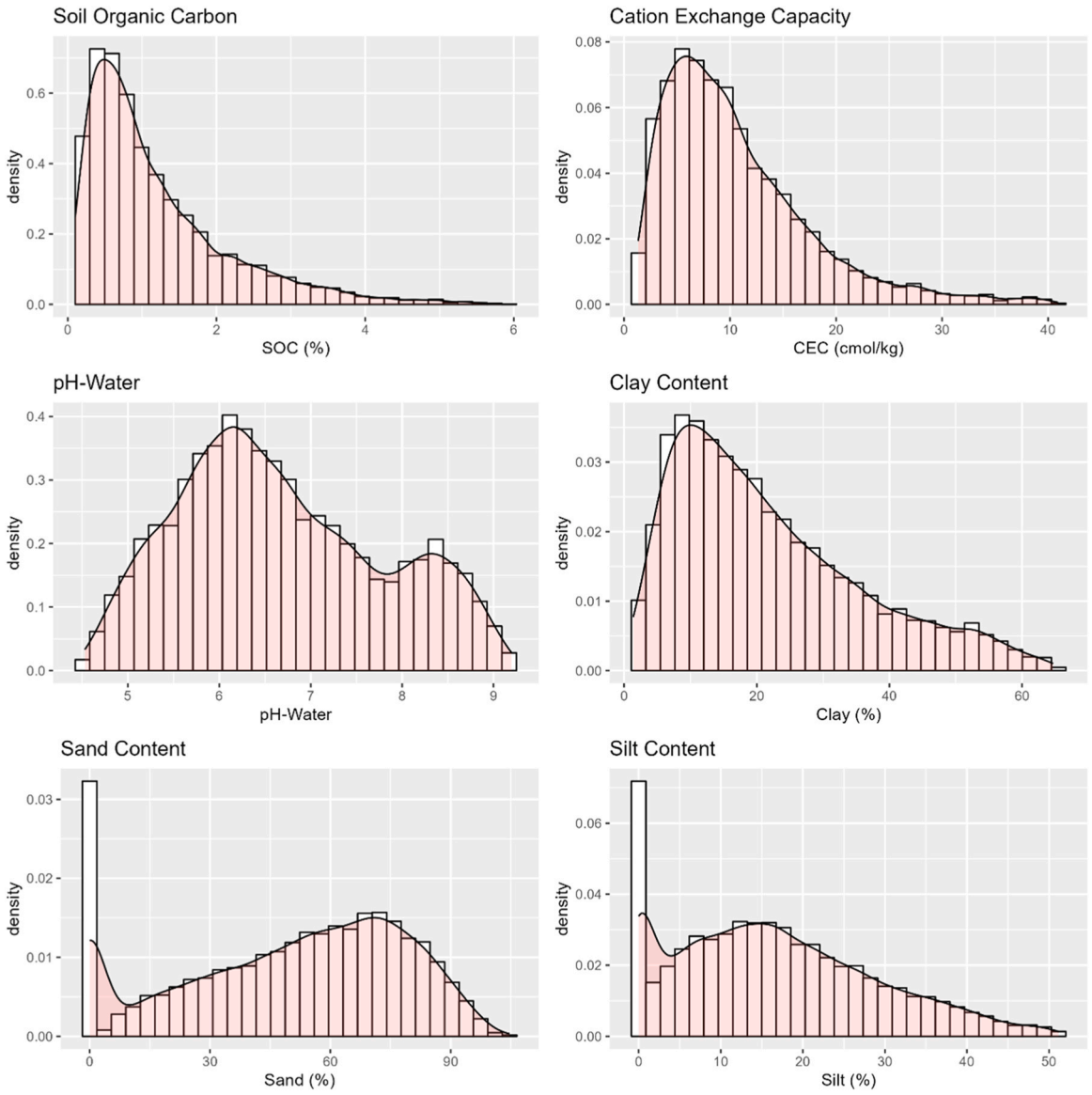


Fig. 2. Distribution of the measured soil nutrient parameters.

Table 3
Summary of the model performance for each generated GSNmap product.

Variable	ME	MAE	RMSE	R	R^2	MEC
SOC (%)	-0.02	0.46	0.66	0.74	0.55	0.55
Clay (%)	-0.32	8.55	10.93	0.64	0.42	0.41
pH-soil	-0.01	0.54	0.69	0.78	0.61	0.61
Sand (%)	0.45	14.47	19.63	0.71	0.51	0.5
Silt (%)	-0.08	6.87	8.96	0.68	0.46	0.46
CEC (cmol(+)/kg)	-0.19	4.16	5.63	0.6	0.36	0.35

Lemercier et al., 2022), food security, environmental degradation, climate change, and other environmental domains. Even though other global DSM products exist (such as the Innovative Solutions for Decision Agriculture Ltd-iSDA (Hengl et al., 2021) that include South Africa, it is necessary to constrain the models to a local setting as to limit the impact of other datasets from other regions that may possibly increase variance. Products such as iSDA are very useful for global applications but may not necessarily reflect local variations due to the generalized nature of the model used. The *GlobalSoilMap* project promotes locals to generate their own country-level DSM products, by doing so developing local expertise in the process, which otherwise is difficult to achieve with global products that are usually constrained to international institutions. *GlobalSoilMap* products generated at country level are most likely to be more accurate compared to global products such as the Soil and Terrain Digital Databases (SOTER), iSDA, and AfSoilGrids250m, according to Bodenstein et al. (2022), who evaluated global and continental scale soil maps using point data from the surveys and generally observed poor correlations.

The United States, Australia, France, and Denmark were the among first countries to generate *GlobalSoilMap* products (Arrouays et al., 2014, 2020; Chaney et al., 2019). Since then, countries such as China (Liang et al., 2019), India (Dharumarajan et al., 2019), and some of the African countries like South Africa and Kenya have since joined the *GlobalSoilMap* initiative. This significant progress in DMS products at a national scale is very critical in addressing challenges related to agriculture and climate change. Chen et al. (2020) used a five-model averaging approach to improve the SOC map of France. The authors found that the Bias-corrected Variance Weighted (BC-VW) approach performed better as it consistently gave lower RMSE values with different sample size configurations. Other countries such as Madagascar, Denmark, Nepal, and Tanzania have developed topsoil SOC maps to better understand

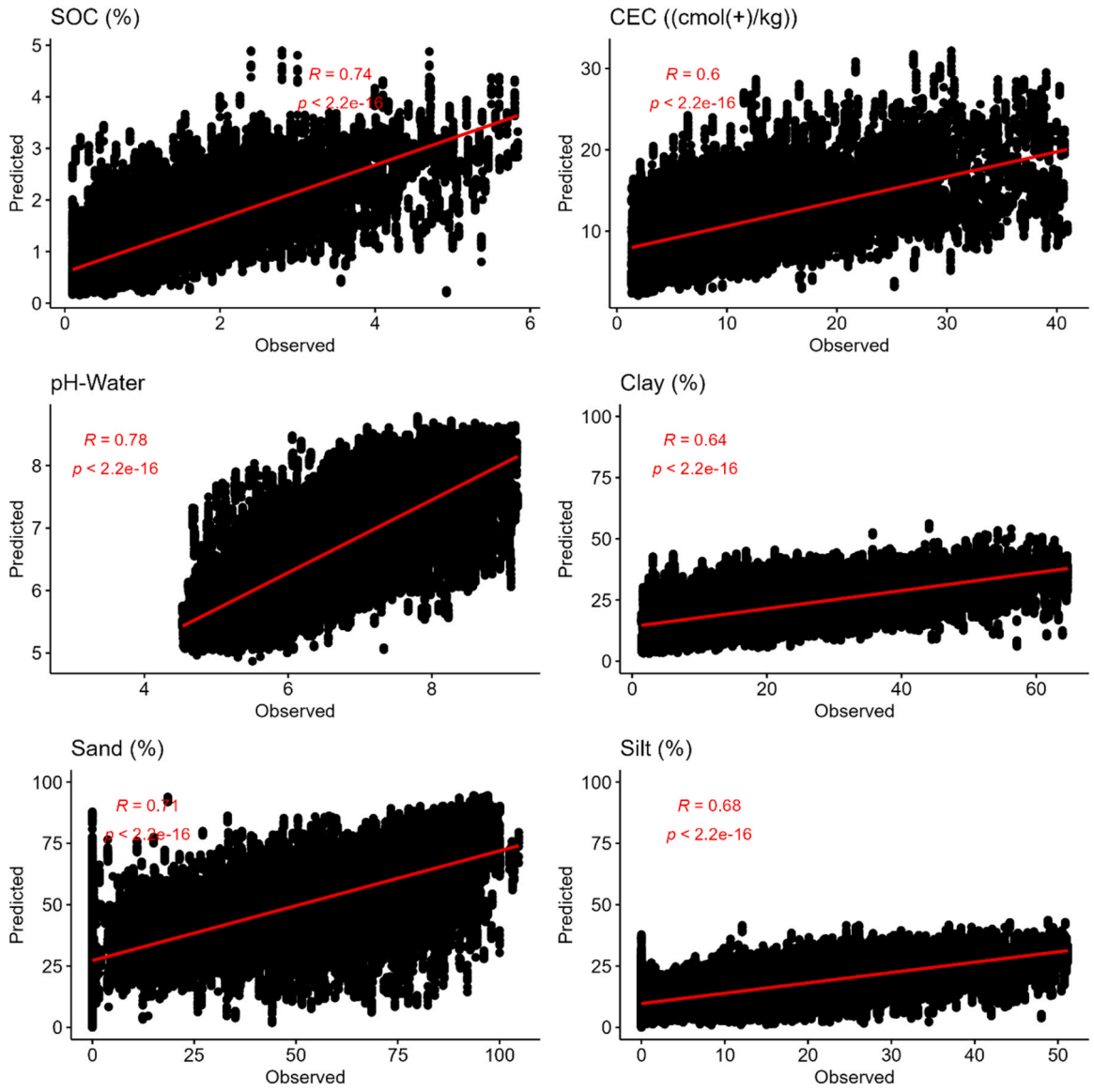


Fig. 3. Scatter plot illustrating QRF model performance of different soil nutrient parameters.

carbon distribution within the soils as it is estimated that soils can store two to three times more carbon compared to vegetation, making soil one of the most important resources for any country (Chen et al., 2020). Venter et al. (2021) have developed a SOC map for South Africa using static and dynamic biological, climate, and morphological environmental variables at 30 m spatial resolution. The results from this study showed some areas that had a decrease (-0.6 % relative to the long-term mean in Nama Karoo biome) and increase (0.3 % relative to the long-term mean in Grassland biome) in SOC trend estimates suggesting habitat losses in some areas. However, this study generated results using QRF model for the SOC, pH-soil, Clay, Sand, Silt and CEC parameters. SOC and pH-soil outperformed other parameters with 55 % and 61 % prediction accuracy. Contrary, Vägen et al. (2016) achieved 74 % and 88 % for SOC and pH with similar model and datasets for African continent. These differences can be attributed to soil variability, sampling designs, climate, topography and among other factors. Hengl et al. (2021) have generated topsoil nutrient maps for the African continent based on different data sources from different countries and applying random forest and XGBoost machine learning algorithms. They obtained an accuracy between 40 % and 85 % for most nutrient parameters.

The generated relative importance of predictor variables from the QRF model (Figs. 4 and Fig 5) provides valuable information about the key *GlobalSoilMap* parameters. The NDVI, land surface temperature, fraction of photosynthetically active radiation (FPAR) and precipitation were among the top important variables to predict SOC. These variables are related to the vegetation density and productivity in the region, suggesting that topsoil is most likely to store carbon when there is high vegetation density (e.g., grasslands, Savanna, forest, Fynbos) compared to when there is low vegetation density (e.g., Nama Karoo) areas. The presence of vegetation increases SOC mainly due to photosynthesis, litter and root biomass, decomposition due to microorganisms and microbial activities (Ontl and Schulte, 2012; Sun et al., 2024). A similar correlation of SOC and vegetation and climate derived parameters were also observed by other authors such as Venter et al. (2021), Liu et al. (2022) and Lotfollahi et al. (2023).

For CEC, the ability of soils to retain nutrients, was well predicted by vegetation, precipitation and land surface temperature parameters. It is also positively correlated with high SOC and clay soils spatial distribution (maps in Figs. 6 and 9, respectively) and negatively correlated with pH and sand, as illustrated in Figs. 8 and 10 respectively. Khaledian et al., (2017) obtained similar results using soil samples from four different nations under different land use conditions. Ghorbani et al.,

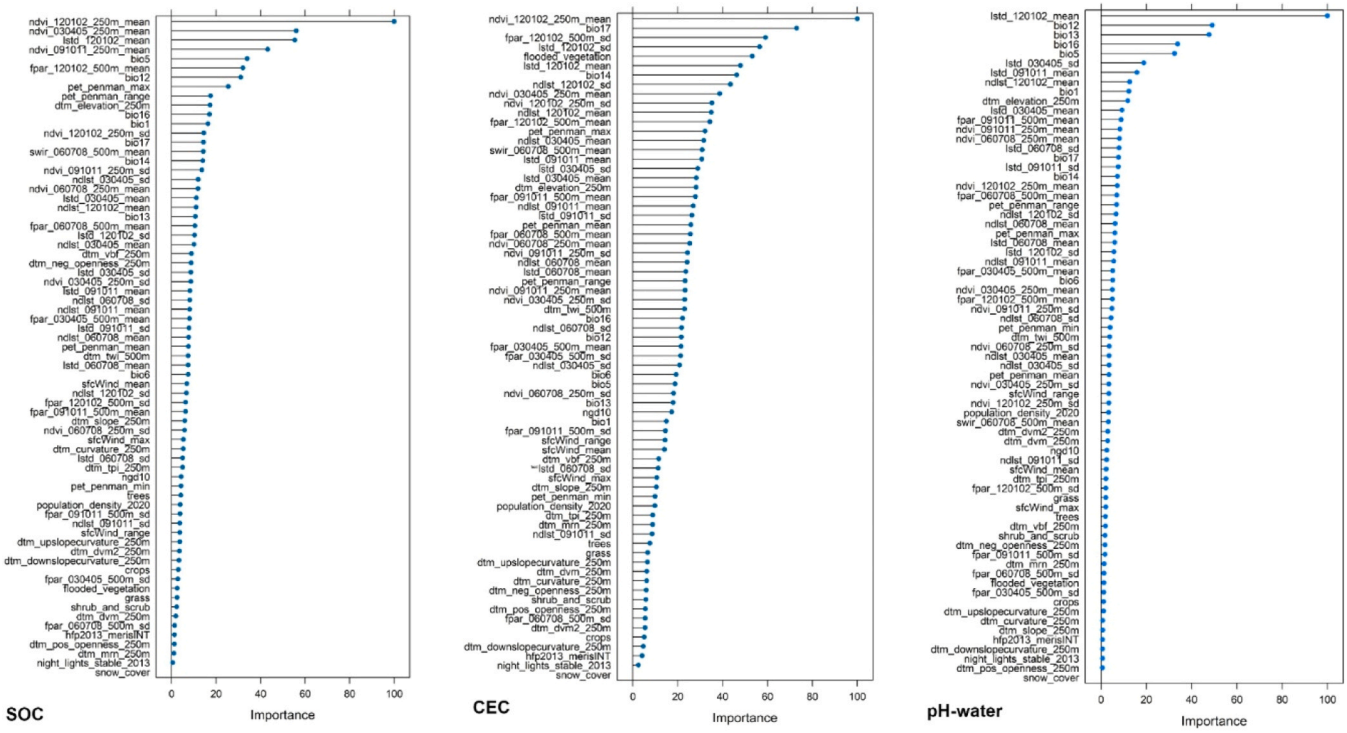


Fig. 4. Variable importance of different soil nutrient parameters (SOC, CEC and pH-soil).

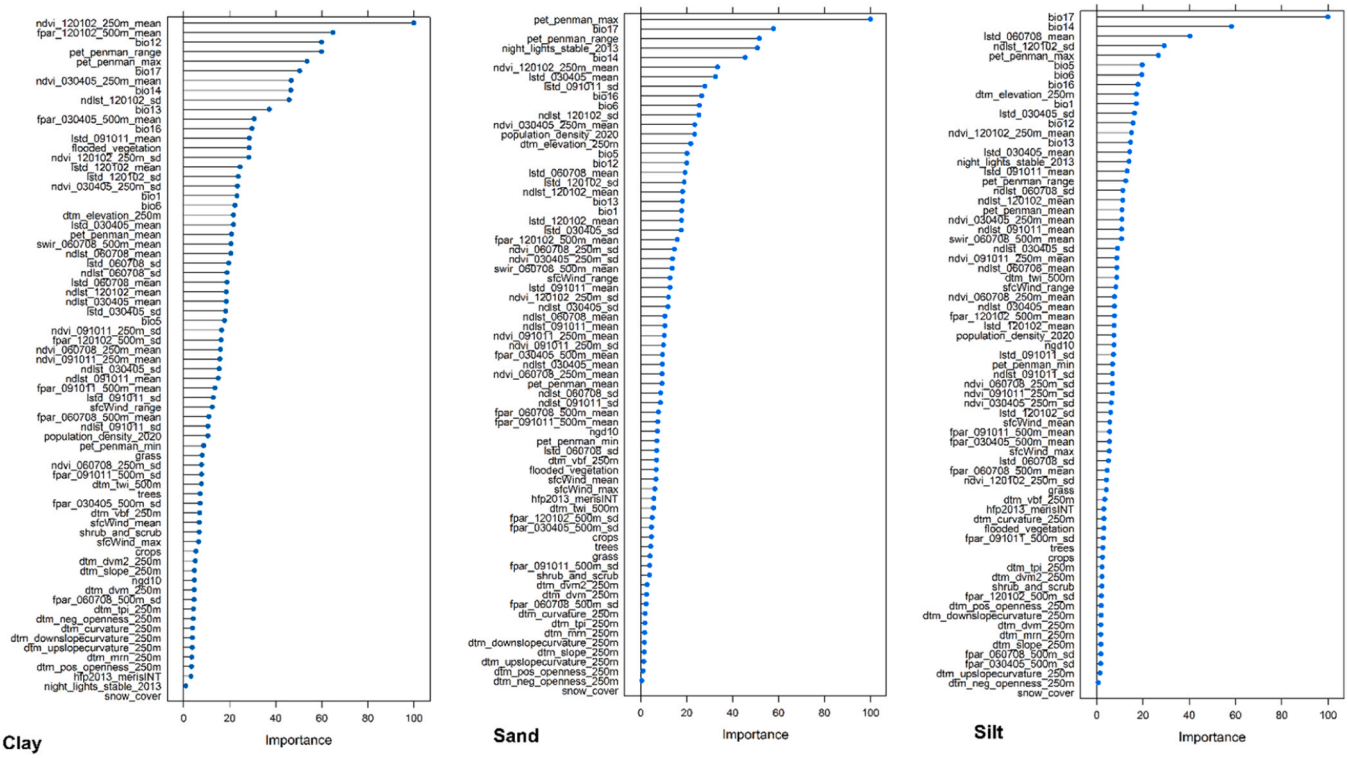


Fig. 5. Variable importance of different soil texture (clay, sand, and silt).

(2015) found that clay and sand had a bigger impact in predicting CEC than silt. Generally, CEC has not been well documented in South Africa, even though it is one of the most critical soil characterization parameters identified in the *GlobalSoilMap* initiative. The soil pH prediction (Figs. 3 and 8) was mostly influenced by land surface temperature, air temperature, and precipitation parameters. Areas with low precipitation e.g., Nama Karoo have high alkaline soils compared to the savanna and grassland biomes. However, it is well known that pH can also be affected by biological processes operating at different scales (Liu et al., 2022). In our study these biological processes are inferred by vegetation activity. For clay, vegetation and precipitation (NDVI, FPAR and bio12) parameters were the most important in the prediction model. Even though, the

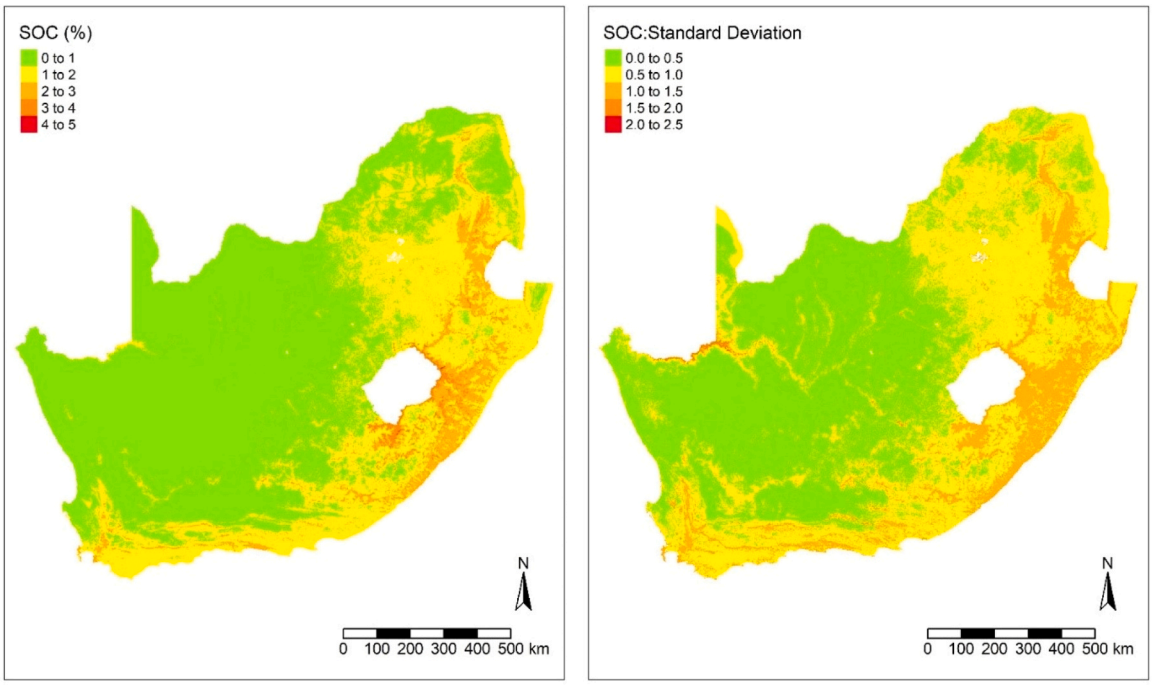


Fig. 6. Spatial variability of soil nutrients: soil organic carbon (SOC) parameter.

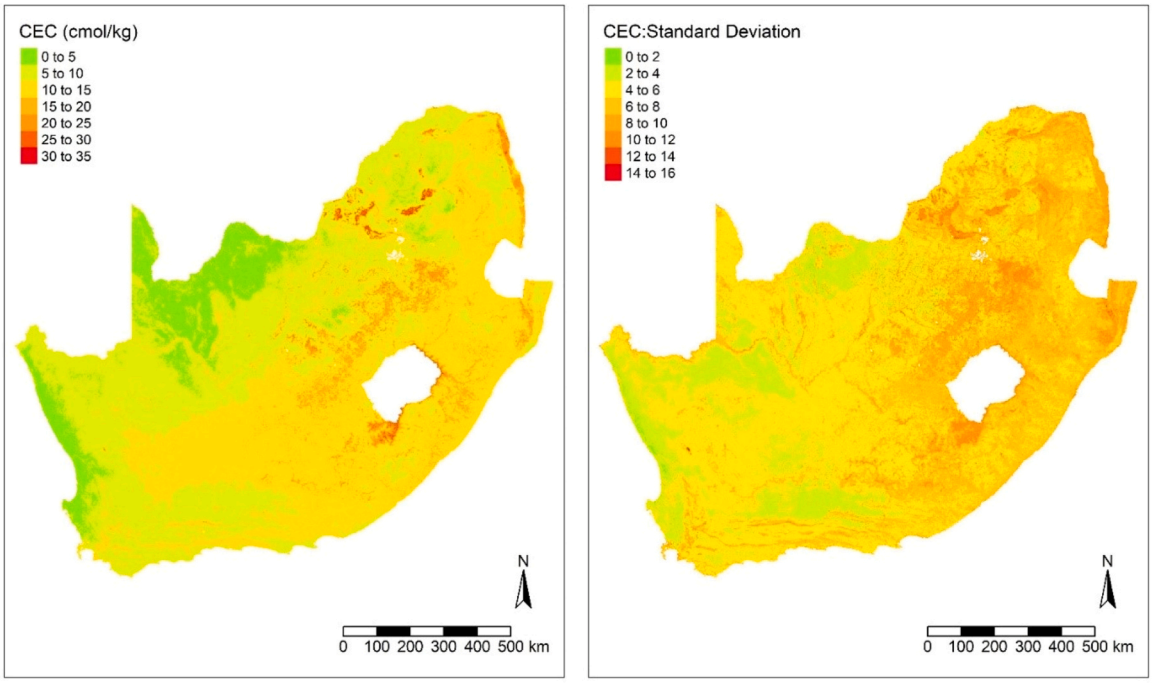


Fig. 7. Spatial variability of soil nutrients: cation-exchange capacity (CEC) parameter.

main factor of clay soil formation is dependent on the erosion and weathering of the bedrock (Ghorbani et al., 2015). Silt and sand were mainly influenced by evapotranspiration and climate parameters.

The generated products can potentially be used in agriculture to understand soil fertility distribution and their changes overtime. Most of the South African population relies on agricultural activities for food production, and this can lead to high demand for fertile soils. Most smallholder farmers do not have resources to carry out detailed soil investigations or to carry out corrective measures. Therefore, these products can be used to assist farmers to re-evaluate site-specific managements zones for farms soil variability over 6 ha to enhance productivity (Marshall and Thenkabail, 2015; Mponela et al., 2020). However, the products generated in this study using MODIS resolution could be constraints to smallholder farmers operating at field and sub-field level because of spatial heterogeneity (Mponela et al., 2020). The products can also be used as input into the mathematical models to enhance crop yield prediction (Liu et al., 2022). South African government has recently signed a Climate Change Bill into law in 2024 (South Africa Climate Change Act, 2024). The act aims to enhance South African climate change response and implement a long-term, just transition

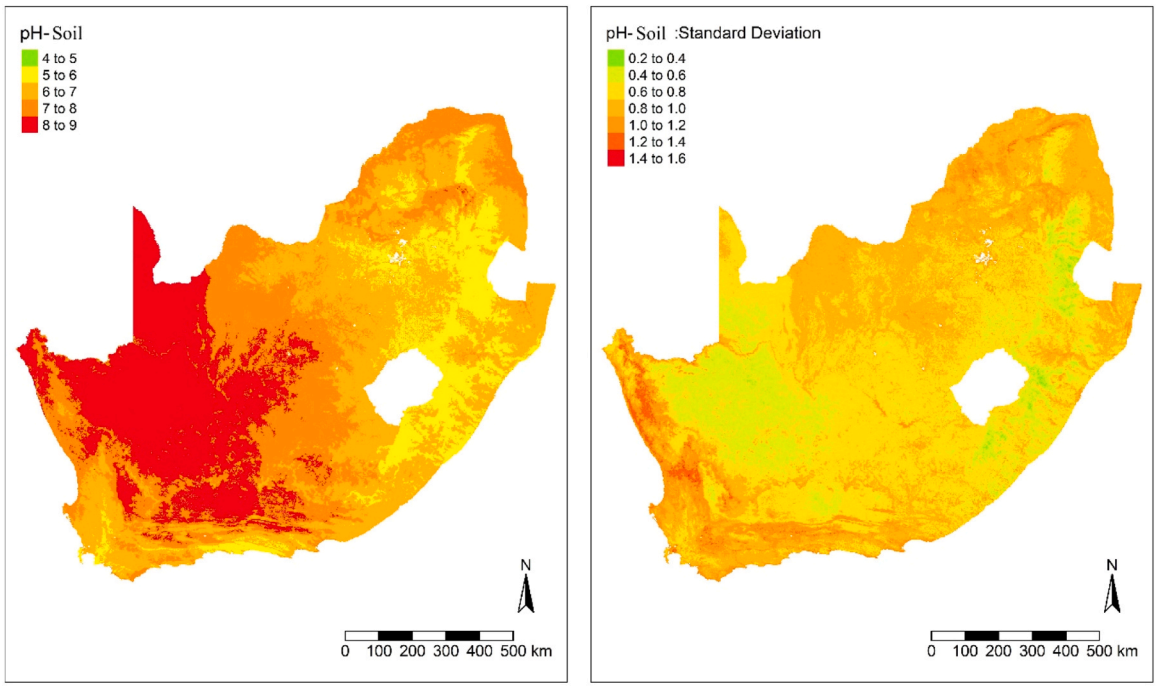


Fig. 8. Spatial variability of soil pH parameter.

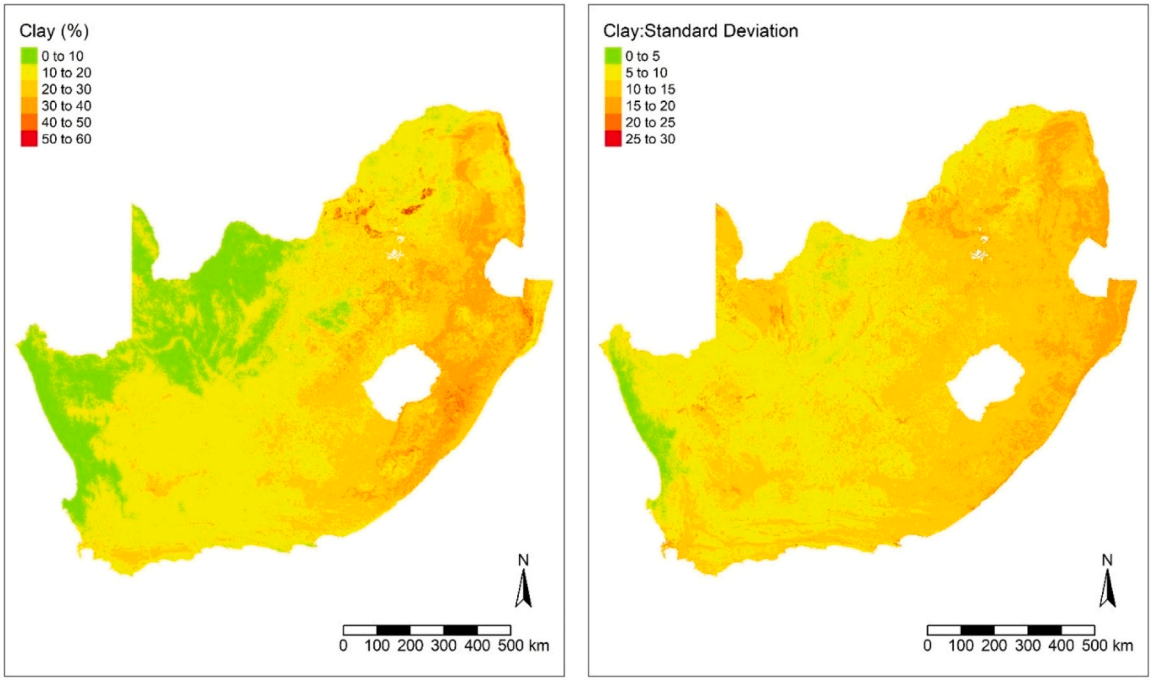


Fig. 9. Spatial variability of clay parameter.

to a low-carbon and climate-resilient economy and society. Products such as SOC can be used to contribute towards low carbon agroecological systems. Soils play a critical role in carbon storage (Dharumarajan et al., 2019; Venter et al., 2021; Lal et al., 2021). Thus, it is critical to understand distribution of soils with high carbon storage capacity.

4.1. Limitation and practical implication on regional-scale digital soil mapping

Our study used other spatially derived datasets such as population density, land cover maps, land surface temperature layers, etc. These spatial layers have inherent errors associated with them. Therefore, it is possible that some of these errors may have propagated into the estimates of *GlobalSoilMap* products. The use of legacy datasets may also introduce bias. This is mainly due to the impact introduced by soil use and management practices, which can potentially affect e.g., pH and SOC over time resulting in errors (Lemercier et al., 2022). Areas in the Karoo have smaller numbers of soil samples compared to the areas in the north, east and south. This can introduce errors in estimating soil parameters in the Karoo regions. Most institutions have a limited data

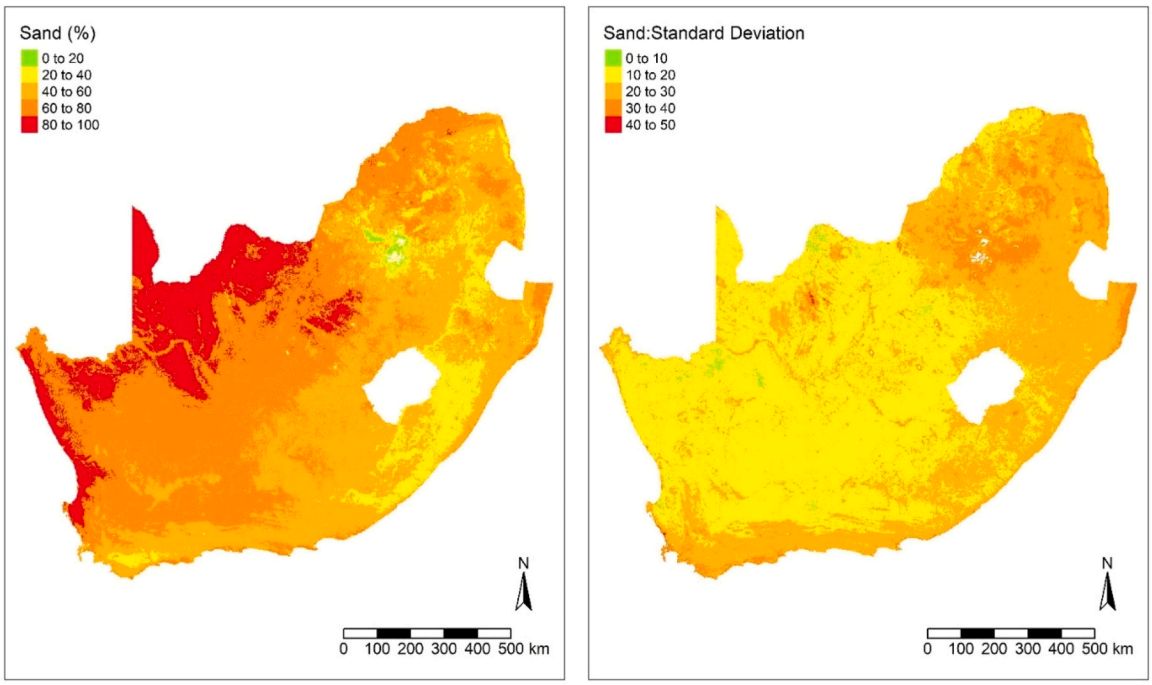


Fig. 10. Spatial variability of sand parameter.

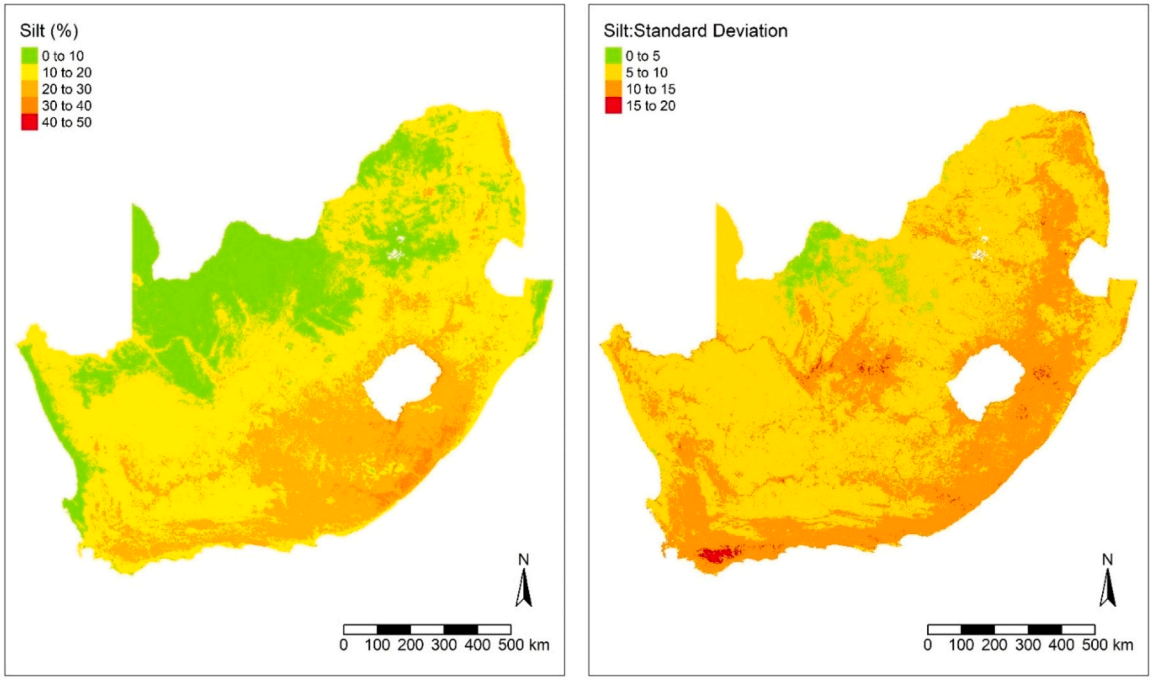


Fig. 11. Spatial variability of silt parameter.

sharing policy, which results in large data gaps within the database. This can be overcome by improving or developing data sharing policies. This initiative can also enhance collaboration of soil scientists within and across the country. The future work of this study will involve mapping of micronutrients. The fine scale GlobalSoilMap project has already been demonstrated by Arrouays et al. (2014), Padarian et al., (2017), and Arrouays et al. (2020). This can be achieved by utilizing integration of Landsat and Sentinel-2 data sets, which have 30m-15m and 10 m spatial resolutions respectively. Variations in depth of the *GlobalSoilMap* products should be included, where possible. More funding is required to improve and maintain the Soil African national soil database.

5. Conclusions

This study contributes towards the GlobalSoilMap project by deriving selected essential parameters together with their uncertainties in South Africa. Maps for sand, clay, silt, CEC, pH-soil and SOC were generated using the method prescribed by the GlobalSoilMap project. The QRF machine learning model was used to predict the soil parameters. Our results indicated the acceptable performance of the QRF model

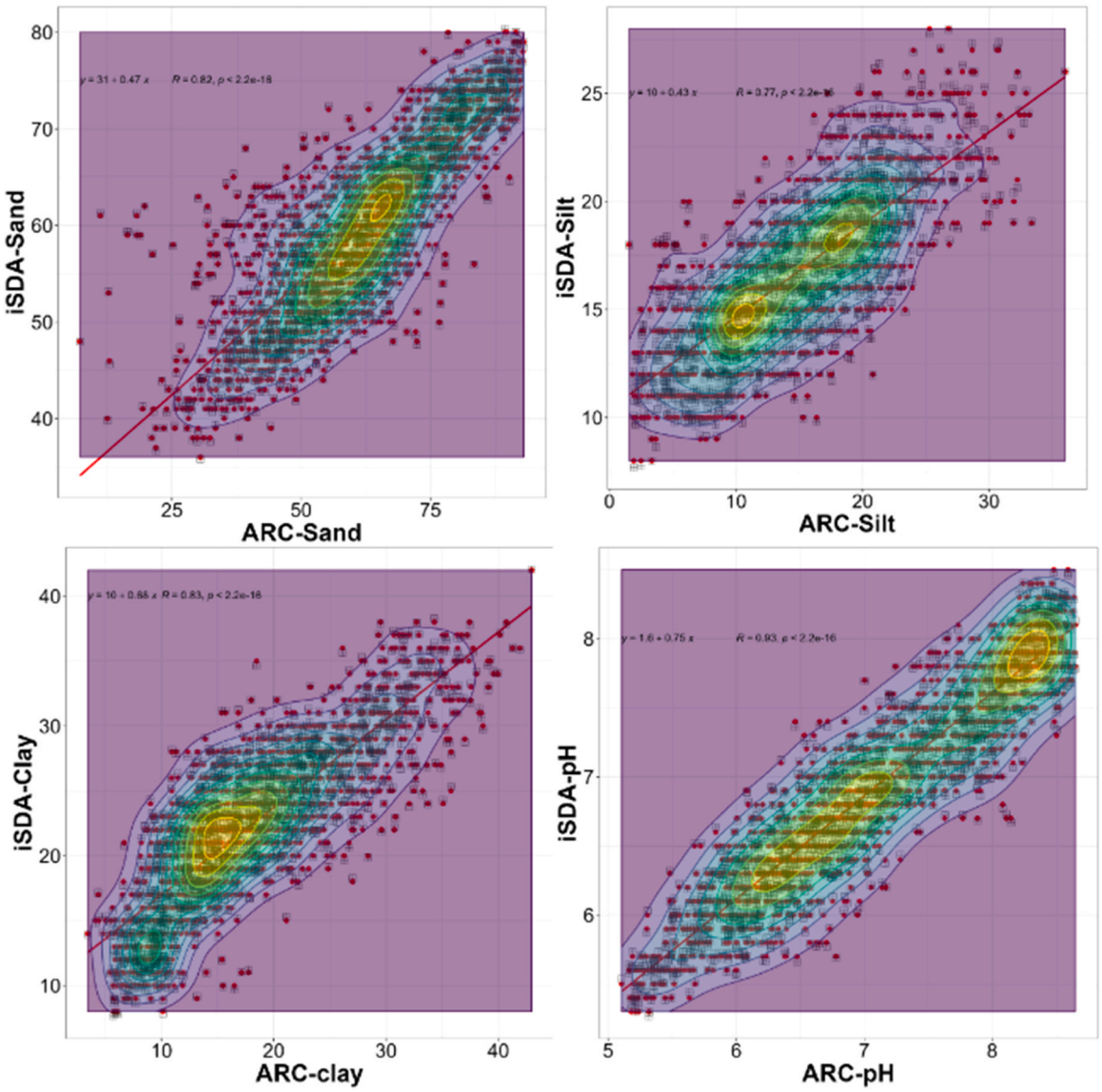


Fig. 12. Comparison of ARC derived soil nutrients with the Innovative Solutions for Decision Agriculture Ltd (iSDA).

in predicting the parameters. The utility of QFR enabled the prediction of mean values and their associated uncertainties. Findings from the study also showed that vegetation indices, land surface temperature and precipitation are important predictors for soil properties. The local maps compared well with global products which confirms the reliability of the local DSMs. Policymakers can benefit by integrating soil nutrient information in their decision-making process to better utilize the available land resources. More work still needs to be done by proving soil nutrient products at different depth levels, including micronutrient mapping. However, financial implications could pose challenges for micronutrient mapping at the country-scale. Outside the current scope of the GlobalSoilMap project, higher resolution datasets such as Sentinel-2 and different processing strategies should be considered in the future.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A1
Summary of soil-forming factors and their proxies of 192 the environmental covariates (source: FAO, 2022)

Factor	Description	Code	Resolution
Temperature	Mean air temperature (annual)	bio1	1000
	Mean daily temperature of warmest month	bio5	1000
	Mean daily temperature of coldest month	bio6	1000
Precipitation	Total precipitation (annual)	bio12	1000
	Mean precipitation of wettest month	bio13	1000
	Mean precipitation of driest month	bio14	1000
	Mean monthly precipitation of wettest quarter	bio16	1000
	Mean monthly precipitation of driest quarter	bio17	1000
Evapotranspiration	Mean monthly PET	pet_penman_mean	1000
	Minimum monthly PET	pet_penman_min	1000
	Range monthly PET	pet_penman_range	1000
	Maximum monthly PET	pet_penman_max	1000
Wind	Minimum monthly wind speed	sfcWind_min	1000
	Maximum monthly wind speed	sfcWind_max	1000
	Range monthly wind speed	sfcWind_range	1000
Growing season	Number of days with mean daily air temperature > 10 °C	ngd10	1000
Vegetation Indices	NDVI (MOD13Q1), mean	ndvi_030405_250m_mean	250
	NDVI (MOD13Q1), mean	ndvi_060708_250m_mean	250
	NDVI (MOD13Q1), mean	ndvi_091011_250m_mean	250
	NDVI (MOD13Q1), mean	ndvi_120102_250m_mean	250
	NDVI (MOD13Q1), standard deviation	ndvi_030405_250m_sd	250
	NDVI (MOD13Q1), standard deviation	ndvi_060708_250m_sd	250
	NDVI (MOD13Q1), standard deviation	ndvi_091011_250m_sd	250
	NDVI (MOD13Q1), standard deviation	ndvi_120102_250m_sd	250
FPAR	Fraction of photosynthetically active radiation (FPAR) (MOD15A2H), mean	fpar_030405_500m_mean	500
	Fraction of photosynthetically active radiation (FPAR) (MOD15A2H), mean	fpar_060708_500m_mean	500
	Fraction of photosynthetically active radiation (FPAR) (MOD15A2H), mean	fpar_091011_500m_mean	500
	Fraction of photosynthetically active radiation (FPAR) (MOD15A2H), mean	fpar_120102_500m_mean	500
	Fraction of photosynthetically active radiation (FPAR) (MOD15A2H), standard deviation	fpar_030405_500m_sd	500
	Fraction of photosynthetically active radiation (FPAR) (MOD15A2H), standard deviation	fpar_060708_500m_sd	500
	Fraction of photosynthetically active radiation (FPAR) (MOD15A2H), standard deviation	fpar_091011_500m_sd	500
	Fraction of photosynthetically active radiation (FPAR) (MOD15A2H), standard deviation	fpar_120102_500m_sd	500
LST	Land Surface Temperature Day (MOD11A2), mean	lstd_030405_mean	1000
	Land Surface Temperature Day (MOD11A2), mean	lstd_060708_mean	1000
	Land Surface Temperature Day (MOD11A2), mean	lstd_091011_mean	1000
	Land Surface Temperature Day (MOD11A2), mean	lstd_120102_mean	1000
	Land Surface Temperature Day (MOD11A2), standard deviation	lstd_030405_sd	1000
	Land Surface Temperature Day (MOD11A2), standard deviation	lstd_060708_sd	1000
	Land Surface Temperature Day (MOD11A2), standard deviation	lstd_091011_sd	1000
	Land Surface Temperature Day (MOD11A2), standard deviation	lstd_120102_sd	1000
NDLST	Normalised Difference between LST Day and LST Night (MOD11A2), mean	ndlst_030405_mean	1000
	Normalised Difference between LST Day and LST Night (MOD11A2), mean	ndlst_060708_mean	1000
	Normalised Difference between LST Day and LST Night (MOD11A2), mean	ndlst_091011_mean	1000
	Normalised Difference between LST Day and LST Night (MOD11A2), mean	ndlst_120102_mean	1000
	Normalised Difference between LST Day and LST Night (MOD11A2), standard deviation	ndlst_030405_sd	1000
	Normalised Difference between LST Day and LST Night (MOD11A2), standard deviation	ndlst_060708_sd	1000
	Normalised Difference between LST Day and LST Night (MOD11A2), standard deviation	ndlst_091011_sd	1000
	Normalised Difference between LST Day and LST Night (MOD11A2), standard deviation	ndlst_120102_sd	1000
SWIR	Black-sky albedo for shortwave broadband (MCD43A3), mean	swir_060708_500m_mean	500
Snow cover	MODIS Snow Cover (MOD10A1) mean	snow_cover	500
Land cover	Dynamic World 10 m near-real-time (NRT) Land Use/Land Cover (LULC) dataset. Mean estimated probability of complete coverage by trees	trees	250
	Dynamic World 10 m near-real-time (NRT) Land Use/Land Cover (LULC) dataset. Mean estimated probability of complete coverage by shrub and scrub	shrub_and_scrub	250
	Dynamic World 10 m near-real-time (NRT) Land Use/Land Cover (LULC) dataset. Mean estimated probability of complete coverage by flooded vegetation	flooded_vegetation	250
	Dynamic World 10 m near-real-time (NRT) Land Use/Land Cover (LULC) dataset. Mean estimated probability of complete coverage by grass	grass	250
	Dynamic World 10 m near-real-time (NRT) Land Use/Land Cover (LULC) dataset. Mean estimated probability of complete coverage by bare	crop	250
Terrain	Profile curvature	curvature	250
	Downslope curvature	downslopecurvature	250
	Upslope curvature	upslopecurvature	250
	Deviation from Mean Value	dvm	250
	Deviation from Mean Value	dvm2	250
	Elevation	elevation	250
	Melton Ruggedness Number	mrn	250
	Negative openness	neg-openness	250
	Positive openness	por-openness	250
	Slope	slope	250
	Topographic position index	tpi	250

(*continued on next page*)

Table A1 (*continued*)

Factor	Description	Code	Resolution
	Terrain wetness index	twi	250
	Multiresolution of valley bottom flatness	vbf	250
Population	Global terrestrial Human Footprint dataset from 2013	hfp2013_merisINT	1000
	DMSP-OLS Nighttime Lights Time Series	night_lights_stable_2013	1000
	Global Human Settlement Layer 2020	population_density_2020	1000

Data availability

Covariates datasets used in the study are available at Google Earth Engine (GEE) platform.

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